

BT WebWorld is a new service from BT, launched in October 1996. It delivers a range of Internet Web-hosting services based on centrally managed webserver farms. The service offers customers cost-effective, powerful, flexible and robust web-server capability and associated support services. BT WebWorld leaves customers free to focus on the opportunities offered by the Internet, with BT handling the complex technical and operational issues.

1. Introduction

This paper outlines the background to BT's decision to offer Web-hosting services, presenting some of the reasons why businesses now account for the majority of Web growth. It then describes the features and benefits offered by the product and describes the service architecture, finally taking a look into the future service possibilities in this exciting and dynamic market.

Once the province of the academic and military communities, the 'Net' is fast becoming a central part of many businesses' communications strategy, and increasingly touches the lives of ordinary people. The World Wide Web (WWW), or simply the 'Web', with its user-friendly interface, has played the largest part in the popularisation of the Net. The Web offers users a series of interactive pages which can include text, graphics, sounds, images and video. These pages are linked together in a seamless way, allowing users to effortlessly browse and select information or services at the click of a mouse. The tool used to access Web information, known as a 'browser', gathers information from one of the millions of Web pages on the Net. It automatically switches between pages and sites to gather the next page of information.

As the popularity of the Web has grown, so have the number of businesses wishing to seize the potential market opportunities offered by it.

As can be seen from Fig 1, the number of public Web sites is doubling every four months, and business users account for the majority of that growth. At the start of 1997, the Netcraft survey identified over 646 000 public Web sites with unique domain names — over eight times more than the same survey a year earlier. The actual number of public Web sites (many of which do not have their own domain name, e.g. <http://www.btwebworld.mycompany>) could be

ten times as large; add in the private Web sites operating within a company's Intranet — for example, BT has over 1000 such internal sites — and it is clear that the Web is set to have a revolutionary impact on business communications.

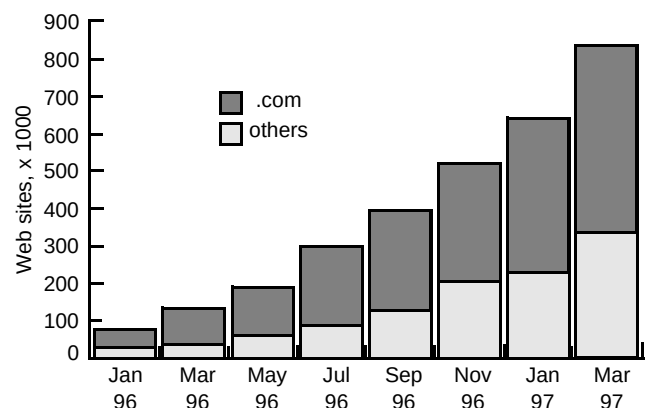


Fig 1 Growth of the Web [1].

1.1 Why are businesses using the Web?

Successful businesses regard the Web not as a network service, but as a market. Buyers and sellers, competitors and partners, products and services, revenue and payment — these are the core elements of the Web market, just like any traditional market. Unlike traditional markets, however, there are very few barriers to entry, it is extremely fast moving and the rewards for the first movers can be great. The Web is being used for communication, research, marketing, customer support, selling and collaborative working (see Fig 2). In many cases, there is a natural

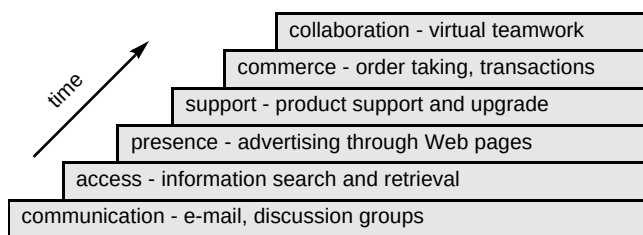


Fig 2 Business use of the Internet is becoming increasingly sophisticated.

progression from basic communication and information access towards increasingly sophisticated uses of the Web. Many businesses achieve payback within months due to cost savings, attraction of new business, speed to market or better market intelligence.

Typical uses of a Web site are wide and varied. Most people immediately think of advertising and selling on an international basis; but companies are finding new applications to exploit the interactivity, availability and multimedia capabilities of the Web all the time. The most common activities include the following.

- Customer services — a Web page can be used to offer 24-hour customer support, 7 days a week. The information given can be far more detailed than is possible by post or even the telephone — examples include the Railtrack on-line journey planner and BT's 'Friends and Family' on-line registration.
- Complementary marketing sites can integrate closely into an organisation's existing marketing activities, such as Vidal Sasoon or the Year of Engineering Success.
- News services — such as the BT Communications Store — offer instant and up-to-date information around the clock on, for instance, partnership announcements, product launches, offers, as well as information resources.
- Product trial and distribution — this is popular with software companies who distribute free trial versions of their products, such as Microsoft and Netscape.
- Micro-marketing — businesses which have specialist products can use a Web site in conjunction with Internet and other advertising to reach a global market of specialist buyers in the most cost-effective way; there are Web sites specialising in everything from Cattle Genetics to Marine Fuel Risk Evaluation.
- Web-zines — Web-based newspapers and magazines (such as the BT Business connections e-zine) can provide a more rounded view of corporate activities, complement a printed edition, or are just published for their own sake.
- Complementary telemarketing — businesses can use a Web site to complement 0800 reception desks; this means that fewer people are needed to answer the phones, and that when customers do call in they are more likely to buy.
- Market research — Web sites can be used to get feedback from customers, suppliers or even employees.
- Collaboration — teams that are based remotely can use a Web page to interact regardless of time zones or locations and are able to meet tighter deadlines.
- On-line product catalogues for retail, mail order, and telemarketing-based companies — with the introduction of multimedia on the Web, product demonstrations are also possible (examples include QualSoft Software and Eurovoice).
- On-line purchasing — it is now possible to sell products and services over the Web using credit card payment, electronic cash and other payment mechanisms (examples include the BT Shop and Tesco).

1.2 External Web hosting

As the Web increasingly becomes a core element of business strategy, so the task of 'hosting' Web content has become mission critical. Few companies, however, have the resource, time or money to build their Web site entirely in-house. Indeed, independent research by Forrester Research Inc suggests that the cost of setting up an internal Web site can be \$221,000 in the first year. For this reason, many businesses choose to outsource their Web hosting to network operators, Internet service providers and equipment vendors, which according to Forrester can slash costs by 80%. Although such cost savings are an important factor, it is not the primary reason for outsourcing Web hosting. Forrester's research shows that the need for external labour and knowledge, increased security, increased speed to market and better service and support were all seen as more important in the decision to outsource.

2. BT WebWorld

Recognising the clear customer demand for Web hosting and related services, BT took the decision to offer a service which would revolutionise the implementation and growth of World Wide Web services. Known as BT WebWorld™, the service combines BT's extensive communications network and support infrastructure with world-class webserver software and computing platforms. By mounting Web applications in a secure, networked environment developed by BT, customers are free to concentrate on supporting their own business objectives, leaving BT to take on the complex technical and operational

activities on their behalf. The service is designed to be as simple or as complex as the customer wants. BT can arrange for complete Web site design and implementation, using a team of accredited Web Partners, or the customers can make use of free tools and on-line support to build their site themselves. For the more advanced user, the service includes a range of sophisticated features and a built-in library of functions.

Using BT WebWorld, businesses can offer a fast and reliable service to their customers at a fraction of the cost of hosting their own. Customers can have a Web site up and running within a day or so, and, with low monthly costs, few businesses will find difficulty in entering the market. The product group comprises four main offerings.

- **Standard publisher**

This is a shared hosting service which offers an inexpensive entry into the Web publishing arena. Aimed at businesses wishing to establish a simple presence on the Internet with minimum cost and complexity, it includes a range of site-creation tools and access to the Web Partner scheme.

- **Premium publisher**

These 'industrial-strength' shared-hosting Web publishing capabilities are aimed at businesses seeking to achieve a quality presence on the Internet. Built on a high-availability platform, the service delivers defined service-level targets with high availability and throughput.

- **Commerce**

Currently being tested, this service offers on-line credit card purchasing services from BT or third party Web sites. The service is based on Open Market's OM Transact service [2].

- **Dedicated server**

With stand-alone servers for customers wishing to house a customised server application, this service offers a managed server platform to which the customer has full system access.

The benefits of the BT WebWorld service include:

- simple pricing,
- rapid service deployment,
- rapid expansion,
- defined performance levels,
- reduced set-up costs,
- reduced technical risk,
- 24-hour customer service,
- high-bandwidth network connectivity,
- high availability,
- 24-hour operations and maintenance,
- security monitoring,
- service-creation tools and on-line tutorial,
- built-in advanced functionality,
- consulting support from Web Partners.

3. BT WebWorld architecture

The top-level architecture of the BT WebWorld publisher, dedicated and commerce platforms is shown in Fig 3.

Content providers' Web pages are loaded on to web-server hardware running Netscape Enterprise server software and BT application software which delivers the

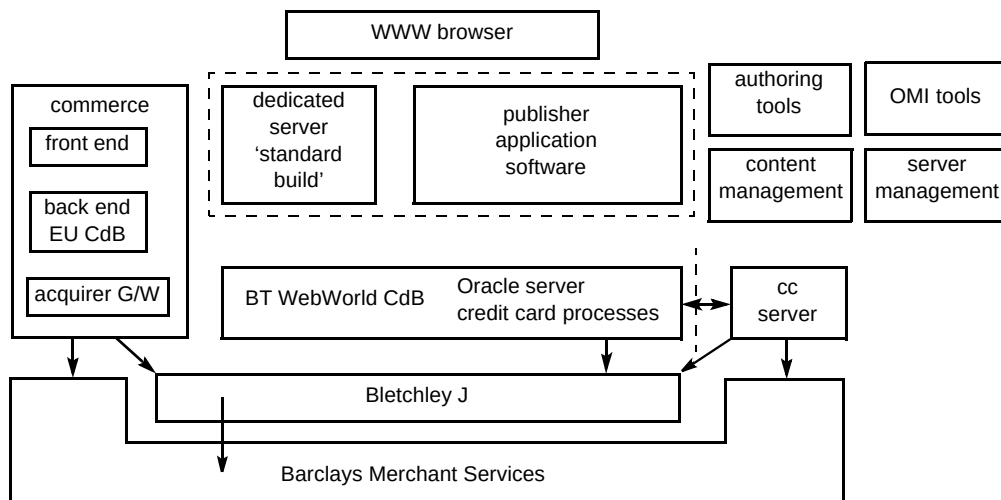


Fig 3 BT WebWorld top-level architecture.

product features. The BT application software interacts with the underlying BT WebWorld Oracle database which contains content provider details. BT WebWorld publisher customers can register on-line, in which case credit card authorisation is obtained via Barclays Merchant Services who act as the merchant acquirer for this product. Alternatively customers can be invoiced. Additionally, content providers can opt to commerce-enable their content using the commerce platform (at the left of Fig 3) which facilitates credit card transactions over the Internet, otherwise known as 'e-commerce' [3].

3.1 Standard and premium publisher platform

Figure 4 depicts the BT WebWorld publisher architecture. This shared Web-hosting platform is based upon Sun Sparc 1000 servers in a high-availability configuration, based in the Cardiff server farm. The publisher application software consists of BT application

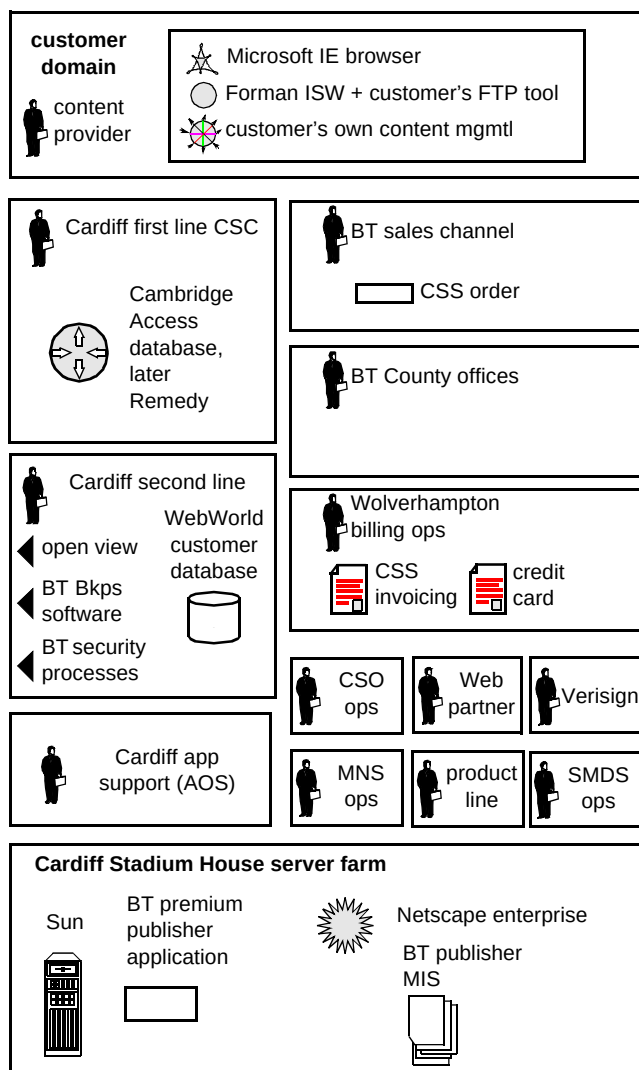


Fig 4 BT WebWorld publisher architecture.

software residing upon Netscape Enterprise server software. The Netscape Enterprise server module provides basic World Wide Web page-serving functionality. The BT application software builds upon this module to provide product features and basic customer account management, including order/registration, create/delete/modify customer account, suspend/enable customer account and reset password. Content providers are advised by e-mail of available and used disk space and traffic quota breaches. Additionally, closed user group (CUG) management is provided via a WWW interface to CUG functions of the Netscape Enterprise server module such that content providers can specify which end users can access their content. A set of popular common-gateway-interface (CGI) scripts [4] are provided for use by content providers, including a form handler and page counter. Traffic statistics relating to content provider's Web sites are made available and can be viewed on-line via Analog and IISstat software packages to show statistics such as total traffic delivered, 'hits' per page and destination of each packet. A search engine provided by the service can be used to index Web sites.

Content providers can opt to have secure socket layer (SSL) security [5], in which case certification is arranged with Verisign and digital certificates are installed within the Netscape Enterprise server. Also they can opt for hardware or software virtual hosting for 'vanity' Web sites. RealAudio is supported via a RealAudio player installed on the server. E-mail forwarding can be configured using Nexor software and domain name service (DNS) configuration whereby e-mail associated with a content provider's domain name (e.g. sales@companyx.com) can be redirected to a remote mailbox.

For billing purposes, invoicing is implemented via BT's telephony billing and customer service system (CSS). Additionally, the latest release of the BT application software interacts with the BT WebWorld customer database via an application programming interface (API) to facilitate credit card billing and on-line registration. Software processes accompanying the database facilitate credit card billing and reporting. A credit card authorisation module performs an authorisation with the acquirer (Barclays Merchant Services). Subsequently, settlement files are sent via BT's internal networks to Barclays Merchant Services.

3.2 Dedicated server platforms

Dedicated servers are particularly appropriate for high-value content providers who do not 'fit' well on BT WebWorld publisher for technical or commercial reasons. This can happen when they do not wish to share a server with other customers, have requirements for extensive CGI

scripting or database support, have high-volume traffic requirements, or demand NT platforms. In these cases the dedicated-server product provides them with high performance webserver infrastructure tailored to meet their needs.

Customers of the dedicated server product typically have the most challenging requirements for Web hosting. Dedicated servers are usually located in Telehouse (the site of the UK's primary Internet exchange) so as to obtain optimal network connectivity. Standard build Unix and NT application software is used to provide a basic building-block platform. Customers then select which webserver, database and other applications software is to be loaded on to the core platform.

3.3 Commerce platform

BT WebWorld commerce enables buyers and merchants to securely buy and sell over the Internet. Buyers can use any Visa or Mastercard credit or debit card for purchases, though additional payment mechanisms such as e-cash, procurement cards and secure electronic transaction (SET) [6] are planned for the future. Merchants are responsible for managing and hosting their Web content.

BT's central transaction service captures the customer's details, authenticates registered users, calculates the total purchase price and carries out on-line credit card authorisation and settlement (see Fig 5).

Buyers and merchants can view order status and account information 24 hours a day, 7 days a week using on-line 'smart statements'. In addition, buyers can register on-line to speed up the buying process for all merchants connected to the WebWorld commerce service [7]. Registered buyers need only enter an account name and password at the time of purchase rather than full name, address and credit card details. The service supports purchase of soft goods (e.g. information or software which can be downloaded over the Internet), hard goods (goods or services which are delivered through other channels, e.g. by post), and subscriptions (soft goods to which the buyer can subscribe for a period of time, e.g. an on-line publication).

Not all products can be sold effectively over the Internet. Those which are most successful tend to be attractive to the Internet audience (predominantly male, technically aware, between 20 and 35 years old). Figure 6 summarises some of the products which are currently selling well in the business and home buyer markets.

	business buyer	home buyer
hard goods	computing office stationary paper reports component/reorders	computing books music (tape/CDs)
soft goods	journal subscription information (finance, legal, reports) software agency services	entertainment, music, education, hobbies, software, agency services, intermediaries

Fig 6 Successful products for Internet commerce.

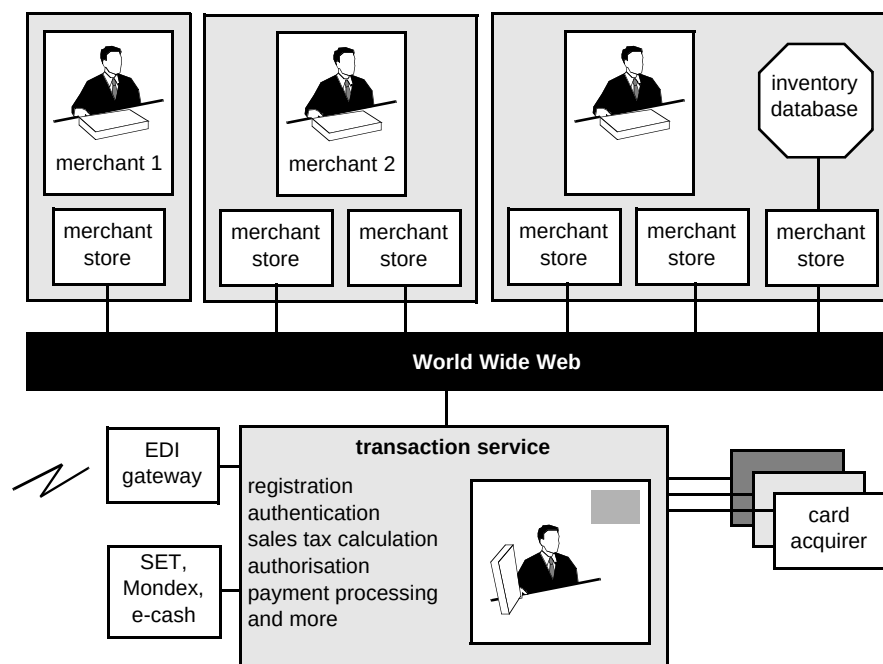


Fig 5 BT WebWorld commerce service.

For the BT WebWorld service, 'commerce enablement' is implemented using Open Market Inc's 'digital offer' technology as shown in Fig 7. This technology intercepts buyers (end users) whenever they attempt to download commerce-enabled content from the WWW server, and requires them to identify themselves and purchase the content before they can access it. Buyers are re-directed to the BT transaction service for on-line registration or card validation, and the card details are passed to the card processor for authorisation and settlement. Once the transaction is authorised, the buyer is issued with a digital receipt which gives them access to the content on the merchant's Web site.

A similar model is used for hard goods purchase and subscriptions. Funds are transferred directly from the buyer's credit/debit card account to the merchant using the merchant's own card processor merchant-ID.

The use of a central transaction server model offers significant improvements in security over in-house solutions. All information transmitted between the buyer and the system (including credit card numbers) is encrypted using the SSL protocol. Credit card details are encrypted on the transaction servers, and are kept encrypted until they are sent to the credit card company over a private network. Indeed, merchants never see the credit card numbers and buyers can only update their credit card numbers — they cannot view their numbers. The user name and password of registered buyers is supplemented by a 'custom challenge' system. This provides a secondary level of security for

transactions above pre-specified limits. Buyers can also set financial limits for purchases made using the system (e.g. only allowing purchases up to £50 in value unless a custom challenge is answered). As only BT handles secure customer data, buyers have no need to worry whether each individual merchant is genuine or has correctly implemented their security procedures.

3.4 Content authoring

For simple HTML content authoring, content providers are able to use the Forman 'Internet Creator' site wizard supplied by BT. This is a simple Windows-based tool which allows customers to create a complete site without any knowledge of HTML or other Web-authoring techniques. The tool features one-button publish capability, making it possible to develop and publish a complete Web site in under an hour. Many customers use commercially available Web-authoring tools to create more advanced sites. Commerce-enabled authoring tools including HotMetal and Bluestone and the low-level 'make digital offer' tool-set are used for the Premium Commerce product.

4. The future

It is challenging to accurately predict the long-term future of the Internet. The main technology trends that will drive new features in BT WebWorld over the next few years are likely to be technology/standards advances, telephony, real-time services and electronic commerce.

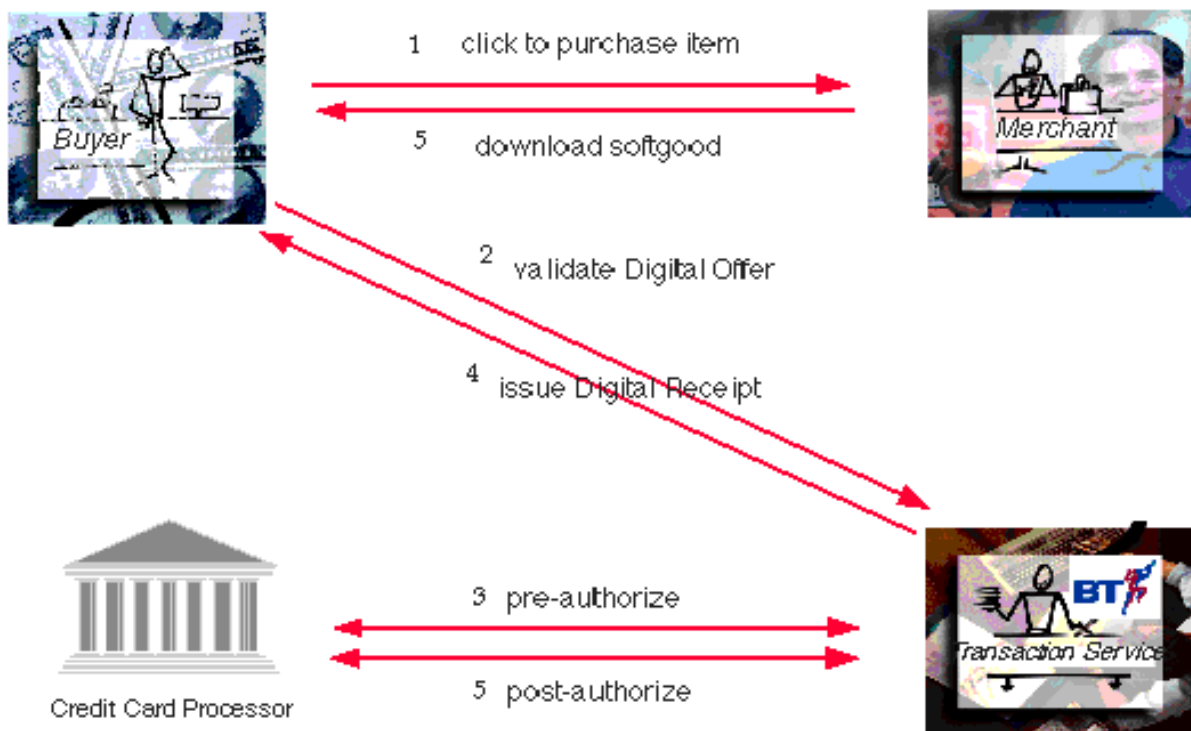


Fig 7 Typical BT WebWorld commerce soft goods purchase flow.

Among technology/standards advances expected to be seen in the near future are intelligent advertising and customer profiling, increased HTML standardisation and a proliferation of companies becoming certification agencies. In the areas of telephony and real-time services, the incorporation of computer/telephony integration with Web pages is anticipated, together with multi-party Web conferencing, widespread real-time telephony support and multicasting. As end-user bandwidth increases, the boundaries between television, radio, telephony and Internet services will become increasingly blurred. The emergence of high-quality real-time video and audio streaming over high-bandwidth multicast networks will create new 'personalised' entertainment and information markets over the Internet. In the area of electronic commerce, WWW implementations of electronic data interchange applications, the introduction of SET technology and mature micro-payment schemes may make the Internet the primary electronic payment infrastructure.

Overall, the Web has moved from academic novelty to business-critical service in little over two years. This is a revolution which seems certain to accelerate, creating significant opportunities for users and equally significant challenges to developers.

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Ian Cameron joined BT in 1981 as a sponsored university student. After graduating with a degree in Electrical Engineering from Imperial College in 1985 he joined BT's research department where he developed interactive speech services, including telephone competitions and telephone banking. In 1991 he joined BT's Product Management division where he led the product launch of new calling card services such as VisaPhone. In November 1993 he became Manager for BT's video on demand trial and more recently has led product development of international

broadband services and groupware services. He is now head of Web-hosting services within BT.



David Patrick joined BT in 1992 after graduating with a PhD in Physics from Imperial College. He initially joined BT's research department where he published a number of papers in the field of ultrafast fibre optics. Subsequently he joined BT's network studies unit where he pursued the topic of Internet evolution.

He currently works within BT in the role of Internet and on-line manager, with technical design authority responsibility for Web-hosting and related content services within BT.